Rigorous proof that a given Hamiltonian dynamical system is ergodic (or mixing) is extremely difficult for even simple cases, and is likely to be dependent on the choice of the potentials, themselves.

In practical simulation, as we cannot consider an infinite t with finite computational resources, experimenters often must choose between estimating $\langle \phi, \rho_\infty \rangle$ using a single long trajectory, or multiple shorter trajectories. Both techniques have their merits, although the correct procedure will be largely system-dependent. For example, let us consider a system with a large number of deep wells (consider an eggbox-like landscape), where our initial condition resides at a peak in-between wells. We may find that when using one single trajectory we only sample the space inside a small number of basins, as the trajectory takes a long time to escape any one well during the simulation. Using multiple trajectories will increase our chance of a thorough sampling of the space, as we are likely to visit more wells in general from our initial condition.

On the other hand, if our initial condition is inside the basin itself, our chances of escaping the well over the course of our simulation is maximized in general by using one long trajectory (simply observe that one trajectory of 100 steps travels further than 100 trajectories of only one step).[7]

It is important to note that a system may be chaotic without being ergodic on a particular level set of energy. For additional discussion of chaotic Hamiltonian systems and their trajectories, see [64, 174, 386].

Example 5.5 (Double Pendulum) To explore some of the issues related to ergodicity, let us return to the double pendulum considered in Chap. 3. For given positive values of the coefficients, the region swept out in the plane by trajectories can be seen to be a union of intersections of pairs of disks centered at the origin and $(0, -2)$, with radii r_1, r_2 respectively, which themselves represent a point on an ellipse in the first quadrant. Determining the boundary of this region analytically is difficult but it is easy to find it numerically. For the special case $k_1 = k_2 = 2$, $\ell_1 = \ell_2 = 1$, and $\mathrm{E} = 4$ that the projected region covered by positions at this energy is the interior of the level set $U = \mathrm{E}$ shown as a dark oval curve in the three panels of Fig. 5.4. If the system were ergodic, we would expect that initial conditions with energy $\mathrm{E} = 4$ would result in trajectories which traverse the outlined region. Instead, we find that the trajectories are generally confined to only an isolated subset of the accessible set. The trajectories starting from different initial conditions would therefore automatically result in different averages of functions that depend on position. The energy surface is three-dimensional (represented as the zero-set of a single constraint in four-dimensional space), but apparently it bifurcates into a collection of isolated components (bounded by two-dimensional closed surfaces). These are none other than the tori predicted by the KAM theory [386]. An illustration of the presence of these tori is obtained by plotting the

[7] Generally, optimizing the molecular dynamics computational cost between "one long" or "many short" is a nontrivial problem which depends on, for example, the availability of parallel computing power, for some discussion see [7, 140, 233].